

LESSON 4.3

PLOTTING, ORDERING, AND COMPARING DECIMALS



LESSON INTRODUCTION

MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.

LEARNING TARGETS

- I can plot and compare rational numbers represented as a decimal or percent.

BENCHMARK COHERENCE

BUILDING ON

MA.5.NSO.1.4 Plot, order, and compare multidigit numbers with decimals up to the thousandths.

WORKING TOWARD

MA.8.NSO.1.2 Plot, order and compare rational and irrational numbers, represented in various forms.

ESSENTIAL QUESTIONS

- How can number lines be used to determine which value(s) are greater or less than others?
- How can statements of inequality comparing rational numbers be interpreted in context?

LESSON NARRATIVE

Students are introduced to plotting and comparing (using $<$, $>$, or $=$) rational numbers represented as decimals and percentages. The lesson begins incorporating some real-world contexts but focuses more on the mathematical context. Students will use number lines, extending their understanding from Lessons 1-2, to compare values using comparison symbols. Connections will also be made to students' prior understanding from Unit 3, where they learned to rewrite positive rational numbers in different forms.

COMPONENT SUMMARY

4.3.1 WARM-UP (~5 MIN.)

Notice and Wonder with temperatures in two cities

4.3.3 COLLABORATION (5-10 MIN.)

Collaboratively compare two negative decimals in a variety of ways

4.3.5 WRAP-UP (~5 MIN.)

Demonstrate understanding of ordering, comparing, and plotting decimals and percentages

4.3.2 GUIDED INSTRUCTION (10-15 MIN.)

Guided learning of comparing and ordering decimals and percentages in context

4.3.4 YOUR TURN! (10 MIN.)

Practice ordering, comparing, and plotting decimals and percentages

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Rational number
- Decimal
- Percent
- Inequality symbol

MATERIALS

- N/A

ADDITIONAL PREPARATION

- (Optional) Find maps and pictures to help students understand the temperatures in different locations in 4.3.1 Warm-Up and 4.3.2 Guided Instruction.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not initially recognize that numbers farther to the left on the number line are smaller values (and vice versa).
- Students may try to use shortcuts (i.e., moving the decimal a specified number of places) when converting values and making comparisons.
- Students may struggle interpreting statements of inequality in context, particularly as they relate to temperatures and race times.

THINGS TO CONSIDER

- There is a number line within 4.3.2 Guided Instruction to provide an opportunity to make the connection that numbers farther to the left are smaller values.
- Use the language “is less than” or “is greater than” rather than “less than” or “greater than” when reading inequality statements. Using the latter can create misconceptions later when students interpret verbal statements and relate them to algebraic statements. For example, “3 less than x ” would be expressed as “ $x - 3$ ” not “ $3 < x$.” If this is not modeled correctly when using inequalities, it will be hard to undo later.

LITERACY AND LANGUAGE ROUTINES

Take Turns

In 4.3.3 Collaboration, students determine which inequality statements are true. Students can take turns explaining, justifying, and asking clarifying questions. The listening partner can also paraphrase their partner’s statements to check for understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.1.1 Participate in Effortful Learning

Students have opportunities to actively participate in effortful learning individually, such as in 4.3.4 Your Turn!, and collectively, such as in 4.3.3 Collaboration. Students must use higher-order thinking skills as they plot, order, and compare decimals. Students can work independently and then with a partner or group to encourage and support active problem-solving.

DIFFERENTIATE INSTRUCTION

Consider showing students the geographical location of states referenced (Alaska, Florida, etc.) on a map and some pictures that show the climate in 4.3.2 Guided Instruction.

4.3.1 WARM-UP: NOTICE AND WONDER

Component Overview: Students share what they notice and wonder about the temperatures in two different cities to apply their understanding of positive and negative numbers from previous lessons.

QUESTIONING

Can you create a number line to compare temperatures? Students can create a vertical number line to display the Memphis and Bangor temperatures. Consider this extension for homework.



4.3.1 Warm-Up: Notice and Wonder

- The diagram gives the temperature, in Fahrenheit (°F) and in Celsius (°C), for Memphis, Tennessee, and Bangor, Maine.



What do you notice about the temperatures?

Answers will vary. Memphis has positive values only. Bangor has both positive and negative values.

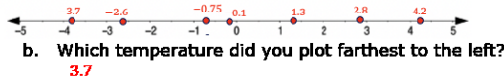


4.3.2 Guided Instruction: Comparing and Ordering Decimals

1. The table shows the lowest temperature, in degrees Celsius ($^{\circ}\text{C}$), for each day during the week of October 14, 2019 in Anchorage, Alaska.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Temperature ($^{\circ}\text{C}$)	4.2	-3.7	-0.75	1.3	2.8	-0.1	-2.6

- a. Plot the temperatures on the number line.



- b. Which temperature did you plot farthest to the left?
3.7

- c. What day of the week is represented by the value that was plotted farthest to the left?

Tuesday

- d. Why does the location of the temperature that is plotted farthest to the left help determine which day had the lowermost temperature?

Answers will vary. Values decrease as you move to the left on the number line.

- e. Interpret the temperature in this context.

During the week of October 14, 2019, the

lowermost temperature was -3.7°C . This was the

☐ warmest
☒ coldest

temperature reached that week.

- f. The inequality symbols, $<$ or $>$, can be used to compare two of the daily low temperatures. Write an inequality that shows the relationship between two of the low temperatures for the week of October 14, 2019.

Answers will vary. Example: $-3.7 < -2.6$

4.3.2 GUIDED INSTRUCTION: COMPARING AND ORDERING DECIMALS

Component Overview: Students compare and order decimals in the context of temperature before extending this to compare decimals and percentages in the context of population.

TEACHER MOVES

Consider showing students the geographical location of states referenced in this component (Alaska, Florida, etc.) on a map and discuss potential differences in the climate.

ENRICHMENT

How can we use a number line to help us determine which value(s) are greater than others? Does this value lie to the left or right of zero on a horizontal number line? Would a vertical number line make sense here? Why or why not?

DIFFERENTIATE INSTRUCTION

Students can role play each scenario, reading each student's response as their own. This strategy provides auditory entry points for students who prefer learning with verbal processing.

2. Jayden, a data analyst, is comparing the populations of four different states to the population of the United States. His data is shown in the table.

State	Population as a Portion of the U.S. Population (decimal form)	Population as a Percentage of the U.S. Population (percent form)
California	0.119	
Florida		6.4%
New York		5.9%
Texas	0.087	

- a. Choose two states to compare. Write or describe your comparison.

Answers will vary. Example: I can compare populations of California and Texas using decimal numbers. Since $0.119 > 0.087$, California has a larger population than Texas.

Example: Florida has a larger population than New York because 6.4% of the US population is more than 5.9%.

- b. Olivia and Anthony's strategies for comparing California and New York are explained below.

Olivia explained, "I know that percent represents the part per 100, so the population of New York is approximately $\frac{6}{100}$ of the population of the U.S. The portion of the U.S. population living in California is approximately 12 hundredths, or $\frac{12}{100}$. Therefore, more people live in California than in New York."

Anthony explained, "I can convert 5.9% to a decimal by dividing 5.9 by 100. So 5.9% can also be represented by 0.059. I know 0.119 is greater than 0.059. Therefore, more people live in California than in New York."

4.3.2 GUIDED INSTRUCTION: COMPARING AND ORDERING DECIMALS (CONTINUED)

TEACHER MOVES

Consider prompting students to rewrite one of the values in each inequality so that both values are in the same form as an additional scaffold. Model how to do this for the first inequality before releasing the others to students.

Olivia and Anthony both used different methods to come to the same conclusion. Discuss their methods with a partner.

Olivia's Method	Anthony's Method
<i>Answers will vary. Olivia converted the decimal to a percentage by rewriting it as a fraction out of 100 first.</i>	<i>Answers will vary. Anthony converted the percentage to a decimal by dividing the percent by 100 so that the values could be compared.</i>

- c. Complete each inequality statement below to compare the given values.

$$0.119 > 6.4\% \quad 6.4\% < 0.087 \quad 0.087 > 5.9\%$$

- d. In which of the four states does the highest portion of the U.S. population live?

California

- e. In which of the four states does the smallest portion of the U.S. population live?

New York



4.3.3 Collaboration: Interpreting Inequality Statements

An inequality statement is given.

$$-8.7 < -8.4$$

- With your partner, use the inequality statement to discuss which of the following statements are true.
 - ☐ -8.7 is farther to the right on the number line than -8.4 .
 - ☒ -8.7 is farther to the left on the number line than -8.4 .
 - ☒ -8.7 is less than -8.4 .
 - ☐ -8.7 is greater than -8.4 .
 - ☐ -8.4 is less than -8.7 .
 - ☒ -8.4 is greater than -8.7 .
- Which value is farther from 0 on the number line? How is knowing which value is farther from 0 helpful when comparing two negative values?
Answers will vary. -8.7 is farther from zero on a number line. The farther a number is from zero when moving to the left on a number line, the smaller the value of that number.

4.3.3 COLLABORATION: INTERPRETING INEQUALITY STATEMENTS

Component Overview: Students collaborate with a partner or small group to interpret an inequality statement in a variety of ways. Students select all the true statements based on a given inequality.

TEACHER MOVES

Take Turns

Students can take turns explaining, justifying, and asking clarifying questions for each statement in question 1 to determine which statements are true. While monitoring the routine, ask students to paraphrase their partner's statements to ensure active listening and comprehension.

Consider providing sentence stems such as:

"This statement is true/false because..."

DIFFERENTIATE INSTRUCTION

Consider prompting students in need of additional support to use their number lines from Lesson 2 as they process through the problems together.

4.3.4 YOUR TURN!

Component Overview: Students practice plotting and comparing rational numbers.

ENRICHMENT

Between which two integers does the value lie?

To which integer is it closest? How will converting rational numbers in different forms to the same form help me to compare and order two or more numbers accurately?

TEACHER MOVES

Notice that students are not writing symbols, just interpreting them. Consider extending question 4 to include using inequality symbols to prepare students for the 4.3.5 Wrap-Up.



4.3.4 Your Turn!

- Explain why each inequality statement is true or false.
 - $-5.2 < -7.1$
Answers will vary. This is false because both values are negative, but -7.1 is farther to the left on the number line than -5.2 , so it is smaller.
 - $3.1 > 38\%$
Answers will vary. This is true because $3.1 = 310\%$ or $38\% = 0.38$. $3.1 > 0.38$ and $310\% > 38\%$.
 - $-37\% > -25\%$
Answers will vary. This is false because -37% is farther to the left on a number line than -25% , so -37% is smaller than -25% .
 - $-12.371 > -12.58$
Answers will vary. This is true because -12.371 is farther to the right on the number line than -12.58 , so its value is larger.
- Plot each of the following values on the number line. Label each point with its numeric value.

$0.4, -1.5, 12\%, -0.25$

 - Complete the inequality statement to show the relationship between the values.
 $0.4 > 12\% > -0.25 > -1.5$
- Select all the values that are greater than -5 .
 - ☒ 1.324
 - ☐ -6.2
 - ☐ -12.67
 - ☒ 0.231
 - ☐ -1.4
 - ☐ -4.789
- Order these values from least to greatest.
 $0.43, -4.2, 0, 50\%, -1.5, -6.712, 19\%$
 $-6.712, -4.2, -1.5, 0, 19\%, 0.43, 50\%$



4.3.5 Wrap-Up: Ticket Out the Door

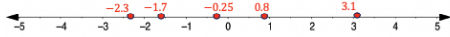
1. Use the symbols $<$, $>$ or $=$ to compare each pair of values.

$$-2.4 < -1.3$$

$$0.5 > 20\%$$

$$-15.21 < -7.136$$

2. Plot the values -2.3 , 3.1 , -0.25 , 0.8 , and -1.7 on the number line. Label each point with its numeric value.



4.3.5 WRAP-UP: TICKET OUT THE DOOR

Component Overview: Students compare values using statements of inequality in addition to plotting each value on a number line.

DIFFERENTIATE INSTRUCTION

The data provided from this formative assessment, coupled with what was observed during the lesson, can help you with differentiating instruction in subsequent lessons.